

Negative Exponent Law

MA.8.NSO.1.3

I can rewrite a power with a negative exponent using a positive exponent, and evaluate it.

Yesterday the exponents stopped at 0.

Today they keep going — and 5^{-2} is not a negative number.

Warm-Up

On your own. Four minutes. No notes.

1. Find the value: $\frac{5^4 \cdot 5^0}{5^2}$
2. Which is greater, $\sqrt{30}$ or 5.5?
3. Solve $-3x = 21$
4. What is the reciprocal of $\frac{3}{4}$? What is the opposite of $\frac{3}{4}$?

Warm-Up

Answers.

1. Find the value: $\frac{5^4 \cdot 5^0}{5^2}$ **25**
2. Which is greater, $\sqrt{30}$ or 5.5? **5.5**
3. Solve $-3x = 21$ **$x = -7$**
4. What is the reciprocal of $\frac{3}{4}$? What is the opposite of $\frac{3}{4}$?
 $\frac{4}{3}$ and $-\frac{3}{4}$

Notes

Copy the table.

I. The pattern keeps going past zero

2^4	2^3	2^2	2^1	2^0	2^{-1}	2^{-2}	2^{-3}
16	8	4	2	1	1/2	1/4	1/8

$$2^{-1} = \frac{1}{2^1}$$

$$2^{-2} = \frac{1}{2^2} = \frac{1}{4}$$

$$2^{-3} = \frac{1}{2^3} = \frac{1}{8}$$

Notes

II. The negative exponent law

$$a^{-n} = \frac{1}{a^n} \qquad \frac{1}{a^{-n}} = a^n \qquad (a \neq 0)$$

$$5^{-2} = \frac{1}{5^2} = \frac{1}{25} \qquad \text{but} \qquad -5^2 = -25 \qquad \text{and} \qquad (-5)^2 = 25$$

A negative exponent means reciprocal. It never means a negative value.

The base keeps its own sign: $(-2)^{-3} = \frac{1}{(-2)^3} = \frac{1}{-8} = -\frac{1}{8}$

Notes

III. The laws still work with negative exponents

$$6^{-2} \cdot 6^5 = \frac{1}{6^2} \cdot 6^5 = \frac{7776}{36} = 216$$

$$\text{or} \quad 6^{-2} \cdot 6^5 = 6^{-2+5} = 6^3 = 216$$

$$\frac{9^{-5}}{9^3} = 9^{-5-3} = 9^{-8} = \frac{1}{9^8}$$

Add, subtract and multiply exponents exactly as before — the exponents are just integers now.

Example 1

Negative exponents top and bottom.

$$\frac{2^{-3}}{9^{-2}}$$

Rewrite with positive exponents, then find the value.

Example 1

Worked.

$$\frac{2^{-3}}{9^{-2}} = \frac{9^2}{2^3} \quad \text{each power swaps sides and loses the minus}$$

$$\frac{9^2}{2^3} = \frac{81}{8} = 10.125 \quad \text{evaluate}$$

Answer: $81/8 = 10.125$

Your Turn

Same steps, your numbers. Boards up when done.

$$\frac{3^{-2}}{5^{-1}}$$

Your Turn

Answer.

$$\frac{3^{-2}}{5^{-1}}$$

swap sides, positive exponents: 5^1 over 3^2

Answer: $\frac{5}{9}$

Example 2

A quotient law crossing zero.

$$\frac{7^0}{7^{-4}}$$

Find the value.

Example 2

Worked both ways.

$$\frac{7^0}{7^{-4}} = 7^{0 - (-4)} = 7^4 \quad \text{quotient of powers: } 0 - (-4) = 4$$

$$\frac{7^0}{7^{-4}} = \frac{1}{7^{-4}} = 7^4 \quad \text{zero exponent, then move } 7^{-4} \text{ up}$$

$$7^4 = 2401 \quad \text{evaluate}$$

Answer: 2,401

Your Turn

Same steps, your numbers. Boards up when done.

$$\frac{(-8)^{-4}}{(-8)^{-6}}$$

Your Turn

Answer.

$$\frac{(-8)^{-4}}{(-8)^{-6}}$$

$$-4 - (-6) = 2, \text{ so } (-8)^2$$

Answer: 64

Whiteboards • Question 1 of 9

Boards up on three.

$$8^{-3}$$

Answer it.

Rewrite with a positive exponent.

Whiteboards • Question 1 of 9

Answer.

$$8^{-3}$$

reciprocal of 8^3

Answer: $\frac{1}{8^3}$

Whiteboards • Question 2 of 9

Boards up on three.

$$2^{-4}$$

Answer it.

Find the value.

Whiteboards • Question 2 of 9

Answer.

$$2^{-4}$$

1 over 2^4

Answer: $\frac{1}{16}$

Whiteboards • Question 3 of 9

Boards up on three.

$$\frac{1}{10^{-2}}$$

Answer it.

Rewrite with a positive exponent, then find the value.

Whiteboards • Question 3 of 9

Answer.

$$\frac{1}{10^{-2}}$$

the power moves up: 10^2

Answer: $10^2 = 100$

Whiteboards • Question 4 of 9

Boards up on three.

$$(-2)^{-3}$$

Answer it.

Find the value.

Whiteboards • Question 4 of 9

Answer.

$$(-2)^{-3}$$

$$1 \text{ over } (-2)^3 = 1 \text{ over } -8$$

$$\textbf{Answer: } -\frac{1}{8}$$

Whiteboards • Question 5 of 9

Boards up on three.

$$5^{-2}$$

Which is the value?

A. $1/25$

B. -25

C. -10

D. $1/10$

Answer it.

Whiteboards • Question 5 of 9

Answer.

$$5^{-2}$$

Which is the value?

A. 1/25

B. -25

C. -10

D. 1/10

Answer: A — 1/25

Whiteboards • Question 6 of 9

Boards up on three.

$$6^{-2} \cdot 6^5$$

Which expression is equivalent?

A. 6^3

B. 6^{-7}

C. 6^{-10}

D. 36^3

Answer it.

Whiteboards · Question 6 of 9

Answer.

$$6^{-2} \cdot 6^5$$

Which expression is equivalent?

A. 6^3

B. 6^{-7}

C. 6^{-10}

D. 36^3

Answer: A — 6^3

Whiteboards • Question 7 of 9

Boards up on three.

$$3^{-4} \cdot 3^4$$

Answer it.

Find the value.

Whiteboards · Question 7 of 9

Answer.

$$3^{-4} \cdot 3^4$$
$$3^0$$

Answer: 1

Whiteboards • Question 8 of 9

Boards up on three.

$$\frac{9^{-5}}{9^3}$$

Answer it.

Write it as one power with a positive exponent.

Whiteboards • Question 8 of 9

Answer.

$$\frac{9^{-5}}{9^3}$$

$$9^{-8}$$

Answer: $\frac{1}{9^8}$

Whiteboards • Question 9 of 9

Take your time. Boards up when you have written it.

Ellen says the value of 5^{-2} is -25 .

Explain why she is wrong, and show the correct value.

Write your answer in sentences.

Use the word reciprocal.

Whiteboards • Question 9 of 9

Answer.

Ellen says the value of 5^{-2} is -25 .

Explain why she is wrong, and show the correct value.

reciprocal, not opposite

Answer: $5^{-2} = 1/5^2 = 1/25$

IXL

Last five minutes.

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Stop at a SmartScore of 67 on each skill. Whatever you do not finish is tonight's practice.

Today's skills

Understanding negative exponents — YBB

Evaluate powers with negative exponents — WGS